

CUSTOMER MILESTONE BRIEF

MS #15 - Conflict-Aware Portfolio Factory

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.15.0
Date	2026-08-15
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #15 - Conflict-Aware Portfolio Factory converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Coordinates multiple modernization work orders and detects shared-file, dependency, dataset, and scheduling conflicts before execution.

Why this matters to a customer

Coordinates multiple modernization work orders and detects shared-file, dependency, dataset, and scheduling conflicts before execution.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a deterministic portfolio controller that loads multiple bounded work orders, binds them to the exact knowledge-graph identity, and emits a content-addressed execution plan.
- Added file-scope, graph-scope, graph-dependency and declared-dependency conflict detection with stable serialization decisions and bounded parallel wave scheduling.
- Added explicit low, medium, high and critical risk classification; high-risk work and critical conflicts fail closed until approved by an external human authority.
- Added HMAC-signed, expiring, tamper-evident human approvals bound to the exact plan, required work orders and acknowledged conflicts; agent identities cannot approve portfolios.

- Added wave barriers, parallel cell dispatch, stop-on-failure behavior and composite portfolio run receipts while preserving independent private acceptance gates for every cell.
- Expanded the demo to bounded INTCALC, POSTTRAN and CRESTMT policy surfaces grounded in the existing CardDemo graph without claiming live z/OS equivalence.
- Added versioned portfolio manifest, plan, approval and run-receipt schemas, cross-platform CLI launchers, a read-only Portfolio Control Tower, and adversarial approval/scheduling tests.

How it differs from earlier work

MS #15 builds on MS #14 (Verified Semantic Memory). The earlier milestone established its own bounded capability; this milestone adds conflict-aware portfolio factory while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use conflict-aware portfolio factory as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.15.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications

- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.